12-9-21

Lecture - 6

Chapter-1

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$P_r^n = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

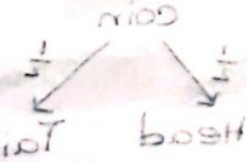
A, B independance –

$$P(A \cap B) = P(A) \cdot P(B)$$

Tree Diagram:

A tree diagram is used to provide a visual representation of probabilities. The outcome of certain event is found at the end of each branch in the tree diagram.

Example:



Scanned with CamScanner

Scanned with CamScanner

Tama Shil
ID: 342

11)

Exercise 01.

Conditional probability

Conditional probability is the probability of one event occurring with some relationship to one or more events.

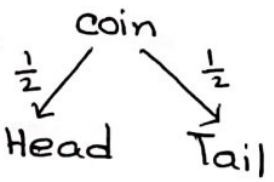
The formula for conditional probability is-

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Tree Diagram:

A tree diagram is used to provide visual representation of probabilities. The outcome of certain event can be found at the end of each branch in the tree diagram.

Example:



Scanned with CamScanner

Scanned with CamScanner

Bayes theorem:

The Bayes theorem is expressed as -

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Proof: where. $P(A)$ = probability of event A

$P(B)$ = probability of event B

$P(A|B)$ = the probability of event A occurring, given event B has occurred.

$P(B|A)$ = the probability of event B occuring, given event B has occurred.

Proof:

we know,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(B)\,P(A|B) \quad \text{——} \ (i)$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A \cap B) = P(A)\,P(B|A) \quad \text{——} \ (ii)$$

from (i), (ii)

$$P(A) \cdot P(B|A) = P(B)\,P(A|B)$$

$$\therefore P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

this equation is known as Bayes theorem.

Scanned with CamScanner

Scanned with CamScanner

Given, $P(B|A) > P(B)$

if $P(B'|A) \not< P(B')$ then

assume

$$P(B'|A) \geq P(B')$$

then, $1 - P(B'|A) \geq 1 - P(B)$

$\Rightarrow -P(B|A) \geq -P(B)$

$\Rightarrow P(B|A) < P(B)$

which contradicts with our initial condition. So, $P(B'|A) \not\geq P(B')$

therefore,

$$P(B'|A) < P(B')$$

(shown).

$$\frac{(B \cap A)\,9}{(B)\,9} \text{ or } = (B|A)\,9$$

(i) —— $(B|A)\,9\,(B)\,9 = (B \cap A)\,9$

$$\frac{(B \cap A)\,9}{(A)\,9} = (A|B)\,9$$

(ii) —— $(A|B)\,9\,(A)\,9 = (B \cap A)\,9$

(ii) . (i) most

$$(B|A)\,9\,(B)\,9 = (A|B)\,9 \cdot (A)\,9$$

$$\frac{(A)\,9\,(A|B)\,9}{(B)\,9} = (B|A)\,9 \quad \therefore$$

Scanned with CamScanner

Scanned with CamScanner

Exercise 02:

i) The probability of the purchaser select a 21-ft^3 = 0.17+0.13 = .30

The probability of the purchaser select a refrigerator with ice-maker

= 0.12+0.17+0.24 = 0.53.

ii) The purchaser select a model with an icemaker, the probability of it was a 21-ft^3 $= \frac{0.17}{0.12+0.17+0.24} = \frac{0.17}{0.53} = 0.321$

The probability of selecting 21-ft^3 unconditionally = 0.17+0.13 = .30

So, we can say that, for selecting 21-ft^3 model, the conditional probability is greater than unconditional probability.

Scanned with CamScanner

iii) The purchaser select a model with at least 20-ft^3 of space.

The probability that the selected refrigerator did not have an icemaker

$$= \frac{0.13 + 0.16}{0.17 + 0.13 + 0.24 + 0.16}$$

$$= \frac{0.29}{0.70}$$

$$= 0.414.$$

Exercise 3:

Let,

S: the event that she will receive a strong recommendation

M: the event that she will receive a moderately good recommendation

W: the event that she will receive a week recommendation

R: the event that she will receive the job offer.

Scanned with CamScanner

So,

$P(S) = 0.7$

$P(M) = 0.2$

$P(W) = 0.1$

$P(R|S) = 0.8$; $P(R'|S) = 1 - 0.8 = 0.2$

$P(R|M) = 0.4$; $P(R'|M) = 1 - 0.4 = 0.6$

$P(R|W) = 0.1$; $P(R'|W) = 1 - 0.1 = 0.9$

a) the event that she will receive the job offer: R

$\therefore P(R) = P(R,S) + P(R,M) + P(R,W)$

$= P(S) \times P(R|S) + P(M) \times P(R|M) + P(W) \times P(R|W)$

$= 0.7 \times 0.8 + 0.2 \times 0.4 + 0.1 \times 0.1$

$= 0.65$

b) $P(S|R)$ Given, she receives the offer, Now, she received a strong recommendation.

$$P(S|R) = \frac{P(R|S) \cdot P(S)}{P(R)}$$

$$= \frac{0.8 \times 0.7}{0.65}$$

$$= 0.862$$

Scanned with CamScanner

she received a moderate recommendation.

$$P(R' P(M|R) = \frac{P(R|M)\ P(M)}{P(R)}$$

$$= \frac{0.4 \times 0.2}{0.65}$$

$$= 0.123$$

She received a week recommendation.

$$P(W|R) = \frac{P(R|W)\ P(W)}{P(R)}$$

$$= \frac{0.1 \times 0.1}{0.65}$$

$$= 0.015$$

-o-

c) Given, she doesnot received the job offers

Now, $P(R') = 1 - P(R) = 1 - 0.65 = 0.35$

she received a strong recommendation.

$$P(S|R') = \frac{P(R'|S) \times P(S)}{P(R')}$$

$$= \frac{0.2 \times 0.7}{0.35}$$

$$= 0.4$$

Scanned with CamScanner

She received a moderate recommendation.

$$P(M|R') = \frac{P(R'|M) \times P(M)}{P(R')}$$

$$= \frac{0.6 \times 0.2}{0.35}$$

$$= 0.343$$

She received a week recommendation,

$$P(W|R') = \frac{P(R'|W) \times P(W)}{P(R')}$$

$$= \frac{0.9 \times 0.1}{0.35}$$

$$= 0.257.$$

—— o ——

Bayes theorem –

Bayes theorem describes the probability of occurance of an event related to any condition. It is also considered for the case of conditional probability.

$$P(E_i|A) = \frac{P(E_i) \, P(A/E_i)}{\sum_{k=1}^{n} P(E_k) \, P(A|E_k)}$$

where $k = 1, 2, 3 \ldots n$

Scanned with CamScanner

$$P(E_i|A) = \frac{P(E_i \cap A)}{P(A)}$$

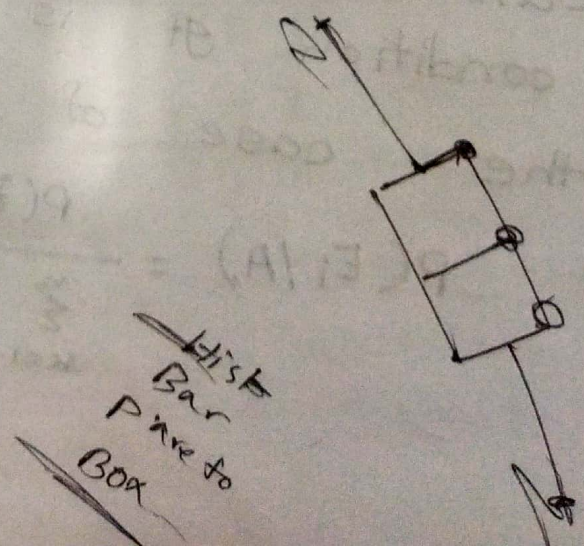
~~$P(E_i \cap A) = P(A)$~~

$$P(A|E_i) = \frac{P(E_i \cap A)}{P(E_i)}$$

$$P(E_i \cap A) = P(E_i)\, P(A|E_i)$$

$$P(A) = \sum_{k=1}^{n} P(E_k)\, P(A|E_k)$$

$$\therefore P(E_i|A) = \frac{P(E_i)\, P(A|E_i)}{\sum_{k=1}^{n} P(E_k)\, P(A|E_k)}$$



Scanned with CamScanner

7-6-21

Lecture -7

PASE 2
chap - 2

Varciable -

i) Discreade variable
ii) Continuous "

X-DRV (discreate random varciable)

P(x) be the probability function or probability mass function

gf. 1. $0 \leq P(X_i) \leq 1$

2. $\sum P(X_i) = 1$

Cumulative Distinction function, $F(x) = P(X \leq x_i) = \sum P(X_i < X$

X	P(X)	$F(X) = \sum P(X_i \leq X)$
1	1/6	1/6
2	1/6	2/6 $(\frac{1}{6}+\frac{1}{6})$
3	1/6	3/6
4	1/6	4/6
5	1/6	5/6
6	1/6	6/6 = 1

Scanned with CamScanner

X - CRV

f(x) - probability density function (Pdf) if-

1) $f(x) \geq 0$

2) $\int f(x)\, dx = 1$

Cumulative distinction function,

$$F(x) = \int_{-\infty}^{\infty} f(x)\, dx$$

(X, P(X)) - prob dist (X, f(x))

* $f(x) = \frac{x}{30}$, $2 \leq x \leq 8$ is f(x) Pdf ?

$$\int_2^8 f(x)\, dx$$

$$= \int_2^8 \frac{x}{30}\, dx$$

$$= \frac{1}{30}\left[\frac{x^2}{2}\right]_2^8$$

$$= \frac{1}{60}(64 - 4)$$

$$= \frac{60}{60} = 1$$

∴ f(x) pdf.

Scanned with CamScanner

$P(X=x_i, Y=y_i)$ be the joint probability function for X and Y DRV

1. $P(X=x_i, Y=y_j) > 0$
2. $\sum_i \sum_j P(X=x_i, Y=y_j) = 1$

$f(x,y) \geq 0$

$\int_x \int_y f(x,y)\, dx\, dy = 1$

Marginal Probability distribution

$$g(x) = \sum_j P(X=x_i, Y=y_j)$$

$$h(y) = \sum_i P(X=x_i, Y=y_j)$$

$$g(x) = \int_y f(x,y)\, dy$$

$$h(y) = \int_x f(x,y)\, dx$$

Scanned with CamScanner

$$f(x,y) = f(x) \cdot f(y)$$

$$P(X = x_i, Y = y_i) = P(X = x_i) \cdot P(Y = y_i)$$

then x, y is independent random variable.

$$g(x|y) = \frac{f(x,y)}{h(y)}$$

$$h(y|x) = \frac{f(x,y)}{g(x)}$$

Scanned with CamScanner

Scanned with CamScanner

Lecture - 8

"chap-2"

14-6-21

The distribution function

The probability of the event $(X \leq x)$ may be expressed as a function of x as,

$$F_X(x) = P_X(X \leq x)$$

Here F_X is called the distribution function or cumulative funct. or CDF. of the random variable X.

Ex- 2.9 →

x	P(X)	$F_X(X)$
0	$\frac{1}{8}$	$\frac{1}{8}$
1	$\frac{3}{8}$	$\frac{4}{8}$
2	$\frac{3}{8}$	$\frac{7}{8}$
3	$\frac{1}{8}$	$\frac{8}{8}=1$

Scanned with CamScanner

If X is a discrete random variable, we associate a number $P_i(x_i) = P_X(X = x_i)$ with each outcome x_i in R_X for $i = 1, 2, \ldots n \ldots$ where

1. $P_X(x_i) \geq 0$
2. $\sum_{i=1}^{\infty} P_X(x_i) = 1$

So, $P_X(x_i) = F_X(x_i) - F_X(x_{i-1})$

$$F_X(x_i) = P_X(X \leq x_i) = \sum_{x \leq x_i} P_X(x)$$

P_X is called the probability function

Tama shil
ID: 342

Exercise 2.9:

Let, X be the number of suits sold.

So, probability mass function,

$$P(X=x) = \frac{e^{-20}\,20^x}{x!} \quad x = 0, 1, 2, \cdots$$

$$= 0, \text{ otherwise}$$

Currently inventory of suits is 30. so, leftover at seasons end is less than or equal to 29.

$$\therefore P(X \le 29) = \sum_{x=0}^{29} \frac{e^{-20}\,20^x}{x!}$$

$$= 0.9782$$

x_i	$P(X=x_i)$	x_i	$P(X=x_i)$	x_i	$P(X=x_i$
0	2.06×10^{-9}	10	5.8×10^{-3}	20	8.8×10^{-2}
1	4.12×10^{-8}	11	1.0×10^{-2}	21	8.5×10^{-2}
2	4.12×10^{-7}	12	1.7×10^{-2}	22	7.6×10^{-2}
3	2.74×10^{-6}	13	2.7×10^{-2}	23	6.6×10^{-2}
4	1.34×10^{-5}	14	3.8×10^{-2}	24	5.6×10^{-2}
5	5.49×10^{-5}	15	5.1×10^{-2}	25	4.4×10^{-2}
6	1.8×10^{-4}	16	6.4×10^{-2}	26	3.4×10^{-2}
7	5.2×10^{-4}	17	7.5×10^{-2}	27	2.5×10^{-2}
8	1.3×10^{-3}	18	0.088	28	1.8×10^{-2}
9	2.9×10^{-3}	19	8.8×10^{-2}	29	1.2×10^{-2}

Scanned with CamScanner

Scanned with CamScanner

So, the probability of suits left over at season's end is 0.9782.

— o —

b)

Exercise 2.10:

$$F_X(x) = 1 - \left(\frac{1}{2}\right)^{x+1} \quad x = 0, 1, 2 \cdots$$

$$= 0 \quad x < 0$$

a) The probability function for X is P_X.

We know

$$P_X(x_i) = F_X(x_i) - F_X(x_{i-1})$$

x_i	$F_X(x_i)$	$P_X(x_i)$
0	$1 - \frac{1}{2} = \frac{1}{2}$	$\frac{1}{2}$
1	$1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}$	$\frac{1}{4}$
2	$1 - \left(\frac{1}{2}\right)^3 = \frac{7}{8}$	$\frac{1}{8}$
3	$1 - \left(\frac{1}{2}\right)^4 = \frac{15}{16}$	$\frac{1}{16}$
4	$1 - \left(\frac{1}{2}\right)^5 = \frac{31}{32}$	$\frac{1}{32}$
5	$\frac{65}{64}$	$\frac{1}{64}$
6	$\frac{127}{128}$	$\frac{1}{128}$
7	$\frac{255}{256}$	$\frac{1}{256}$
8	$\frac{511}{512}$	$\frac{1}{512}$

Scanned with CamScanner

Scanned with CamScanner

b) $P_X(0 < X \leq 8)$

We know,

$$P_X(a < X \leq b) = F_X(b) - F_X(a)$$

$$\therefore P_X(0 < X \leq 8) = F_X(8) - F_X(0)$$

Now,

$$F_X(8) = 1 - \left(\frac{1}{2}\right)^9$$

$$= 1 - \frac{1}{512}$$

$$= \frac{511}{512}$$

$$F_X(0) = 1 - \frac{1}{2}$$

$$= \frac{1}{2}$$

$$\therefore P_X(0 < X \leq 8) = \frac{511}{512} - \frac{1}{2}$$

$$= \frac{511 - 256}{512}$$

$$= \frac{255}{512}.$$

Scanned with CamScanner

Exercise 2.11:

a) $$f_X(x) = kx, \quad 0 \le x < 2$$
$$= k(4-x), \quad 2 \le x \le 4$$
$$= 0, \quad \text{otherwise}$$

a) As, f is a probability density function.

So,
$$\int_0^4 f_X(x)\,dx = 1$$

$$\Rightarrow \int_0^2 kx\,dx + \int_2^4 k(4-x)\,dx = 1$$

$$\Rightarrow k\left[\frac{x^2}{2}\right]_0^2 + k\left[4x - \frac{x^2}{2}\right]_2^4 = 1$$

$$\Rightarrow k(4/2 - 0) + k\left(16 - 8 - \frac{16}{2} + \frac{4}{2}\right) = 1$$

$$\Rightarrow 2k + 2k = 1$$

$$\therefore \quad k = \frac{1}{4}.$$

Scanned with CamScanner

b) As, $k = \frac{1}{4}$,

$$f_x(x) = \frac{x}{4}, \quad 0 \le x < 2$$
$$= \frac{4-x}{4}, \quad 0 \le x \le 4$$
$$= 0, \quad \text{otherwise.}$$

$$\text{Mean } \mu = E(X) = \int_{-\infty}^{\infty} x f_x(x)\, dx$$

$$= \int_{-\infty}^{0} x \cdot 0\, dx + \int_{0}^{2} x \cdot \frac{x}{4} dx + \int_{2}^{4} x \cdot \frac{4-x}{4} dx + \int_{4}^{\infty} x \cdot 0\, dx$$

$$= \frac{1}{4}\int_{0}^{2} x^2 dx + \frac{1}{4}\int_{2}^{4} (4x - x^2) dx$$

$$= \frac{1}{4}\left[\frac{x^3}{3}\right]_0^2 + \frac{1}{4}\left[\frac{4x^2}{2} - \frac{x^3}{3}\right]_2^4$$

$$= \frac{1}{4} \times \frac{8}{3} + \frac{1}{4}\left[2(16-4) - \frac{1}{3}(64-8)\right]$$

$$= \frac{2}{3} + \frac{1}{4}\left(24 - \frac{56}{3}\right)$$

$$= \frac{2}{3} + 6 - \frac{14}{3}$$

$$= \frac{2+18-14}{3}$$

$$= \frac{6}{3}$$

$$= 2$$

Scanned with CamScanner

ciable? What are its type.

$\mu = 2$

$$\text{Variance} = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$$

$$= \int_{-\infty}^{0} (x-\mu)^2 \cdot 0\,dx + \int_{0}^{2} (x-2)^2 \frac{x}{4}\,dx + \int_{2}^{4} (x-2)^2 \left(\frac{4-x}{2}\right) dx + 0$$

$$= \int_0^2 (x^2-4x+4)\frac{x}{4}\,dx + \int_2^4 (x^2-4x+4)\left(\frac{4-x}{4}\right) dx$$

$$= \int_0^2 \frac{1}{4}(x^3-4x^2+4x)\,dx + \int_2^4 \frac{1}{4}(4x^2-16x+16-x^3+4x^2-4x)\,dx$$

$$= \frac{1}{4}\left[\frac{x^4}{4} - \frac{4x^3}{3} + \frac{4x^2}{2}\right]_0^2 + \frac{1}{4}\int_2^4 (8x^2-x^3-20x+16)\,dx$$

$$= \frac{1}{4}\left[4^2 - \frac{32}{3} + 8\right] + \frac{1}{4}\left[\frac{8x^3}{3} - \frac{x^4}{4} - 20\frac{x^2}{2} + 16x\right]_2^4$$

$$= \frac{1}{4}\left(12 - \frac{32}{3}\right) + \frac{1}{4}\left(\frac{512}{3} - 64 - 160 + 64 - \frac{64}{3} + 4 + 40 - 32\right)$$

$$= \frac{1}{4} \times \frac{4}{3} + \frac{1}{4}\left(\frac{512}{3} - \frac{64}{3} - 148\right)$$

$$= \frac{1}{3} + \frac{1}{4}\left(\frac{4}{3}\right)$$

$$= \frac{1}{3} + \frac{21}{3} = \frac{2}{3},$$

Scanned with CamScanner

c)

$$f_x(x) = \begin{cases} \frac{x}{4}, & 0 \le x < 2 \\ \frac{4-x}{4}, & 2 \le x \le 4 \\ 0, & \text{otherwise.} \end{cases}$$

Firstly Let α region

$$F_x(x) = \int_0^{x/2} \frac{x}{4}\,dx$$

$$= \frac{1}{4}\left[\frac{x^2}{2}\right]_0^{x/2}$$

$$= \frac{1}{8}\left[x^2\right]_0^2 = \frac{4}{8} = \frac{1}{2}$$

$$\therefore \int_0^x \frac{x}{4}\,dx = \frac{x^2}{8} \quad \text{---} \quad (1)$$

Secondly let β region,

$$F_x(x) = \int_2^{4x} \frac{4-x}{4}\,dx + \frac{1}{8}\,\frac{1}{2}$$

$$= \frac{1}{4}\left[4x - \frac{x^2}{2}\right]_2^x + \frac{1}{8}2$$

$$= \frac{1}{4}\left(4x - \frac{x^2}{2} - 8 + 2\right) + \frac{1}{8}2$$

$$= x - \frac{x^2}{8} - \frac{6}{4} + \frac{1}{2}$$

$$= x - \frac{x^2}{8} - \frac{3}{2} + \frac{1}{2}$$

$$= x - \frac{x^2}{8} - 1$$

Scanned with CamScanner

$$\therefore F(x) = \begin{cases} = 0, & x < 0 \\ = \frac{x^2}{8}, & 0 \le x < 2 \\ = x - \frac{x^2}{8} - 1, & 2 \le x \le 4 \\ = 1, & x > 4 \end{cases}$$

Exercise 2.21:

$$f_X(x) = \frac{2x}{9}, \quad 0 < x < 3$$
$$= 0, \quad \text{otherwise.}$$

a) CDF →

Firstly

$$F_X(x) = \int_0^x \frac{2x}{9}$$
$$= \frac{2}{9}\left[\frac{x^2}{2}\right]_0^x$$
$$= \frac{x^2}{9}$$

So, $$F_X(x) = \begin{cases} 0, & x \le 0 \\ \frac{x^2}{9}, & 0 < x < 3 \\ 1, & x \ge 3 \end{cases}$$

Scanned with CamScanner

b) Mean $= \mu = E(X)$

$$= \int_{-\infty}^{\infty} x f(x)\, dx$$

$$= \int_{-\infty}^{0} x \cdot 0\, dx + \int_{0}^{3} x \frac{2x}{9} dx + \int_{3}^{\infty} x \cdot 0\, dx$$

$$= \frac{2}{9} \int_{0}^{3} x^2 dx$$

$$= \frac{2}{9} \left[\frac{x^3}{3}\right]_0^3$$

$$= \frac{2}{27} 3^3 = 2$$

Variance,

$$\sigma^2 = \int_{0}^{3} (x-2)^2 \frac{2x}{9} dx$$

$$= \frac{2}{9} \int_{0}^{3} x(x^2 - 4x + 4)\, dx$$

$$= \frac{2}{9} \int_{0}^{3} (x^3 - 4x^2 + 4x)\, dx$$

$$= \frac{2}{9} \left[\frac{x^4}{4} - \frac{4x^3}{3} + \frac{4x^2}{2}\right]_0^3$$

$$= \frac{2}{9} \left[\frac{81}{4} - \frac{108}{3} + \frac{36}{2}\right]$$

$$= \frac{1}{2}.$$

— o —

Scanned with CamScanner

(2) We know,

$$\mu_3' = \int_0^3 x^3 \frac{2}{9} x \, dx$$

$$= \frac{2}{9} \int_0^3 x^4 \, dx$$

$$= \frac{2}{9} \left[\frac{x^5}{5} \right]_0^3$$

$$= \frac{2}{9} \times \frac{3^5}{5}$$

$$= \frac{54}{5}.$$

Scanned with CamScanner

d) ~~Sol~~, Given

$$P_X(X \geq m) = P_X(X \leq m)$$

$$\Rightarrow \int_m^3 \frac{2x}{9} dx = \int_0^m \frac{2x}{9} dx$$

$$\Rightarrow \left[\frac{x^2}{2}\right]_m^3 = \left[\frac{x^2}{2}\right]_0^m$$

$$\Rightarrow 9 - m^2 = m^2$$

$$\Rightarrow 2m^2 = 9$$

$$\Rightarrow m^2 = \frac{9}{2}$$

$$\therefore m = \pm \frac{3}{\sqrt{2}}$$

24)

$$f_X(x) = \frac{1}{\sigma\pi\left\{1 + [(x-\mu)^2/\sigma^2]\right\}}, \quad -\infty < x < \infty$$

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sigma\pi\left\{1 + [(x-\mu)^2/\sigma^2]\right\}} dx$$

Let $\frac{x-\mu}{\sigma} = z$

$$\frac{d}{dz}\left(\frac{x-\mu}{\sigma}\right) = 1$$

$$\frac{dx}{\sigma} = dz, \quad dx = \sigma\, dz$$

Scanned with CamScanner

Scanned with CamScanner

$x \to -\infty, \quad z \to -\infty$

$x \to x \quad z \to x$

Now,

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\pi(1+z^2)}\, dz$$

$$= \frac{1}{\pi}\left[\tan^{-1} z\right]_{-\infty}^{x}$$

$$= \frac{1}{\pi}\left(\tan^{-1} x - \tan^{-1}(-\infty)\right)$$

$$= \frac{\tan^{-1} x}{\pi} + \frac{\pi}{2\pi}$$

$$= \frac{\tan^{-1} x}{\pi} + \frac{1}{2}$$

$$\therefore \quad F_X(x) = \frac{\tan^{-1} x}{\pi} + \frac{1}{2}, \quad -\infty < x < \infty$$

—o—

Scanned with CamScanner

Scanned with CamScanner